Galactic Rotation Without Dark Matter from Elastic Membrane Cosmology:

The Geometric Mean Law and Observational Tests

Shinobu Sakaguchi

Sakaguchi Seimensho (Independent Research), Shisō, Hyogo, Japan

sakaguchi-physics.com

April 2026 (v4.7.8, 2026-04-21 affiliation rev.)

Abstract

We present a membrane cosmology framework in which gravitational strength is modulated by the folding state of a higher-dimensional elastic membrane, producing an effective gravitational acceleration $g_{\rm obs} = (g_N + \sqrt{g_N^2 + 4g_c g_N})/2$ that accounts for observed galactic rotation curves without invoking dark matter. The central result is the *geometric mean law*: the critical acceleration $g_c = \eta\sqrt{a_0 G\Sigma_0}$ is determined as the exact geometric mean ($\alpha = 0.5$) of the universal MOND scale a_0 and the local dynamical surface density $G\Sigma_0$, established from 175 SPARC galaxies with ΔAIC $= -130$ versus MOND and confirmed by extended jackknife resampling. The Condition 15 final form $g_c = 0.584\,\Upsilon_d^{-0.361}\sqrt{a_0\, v_{\rm flat}^2/h_R}$ achieves $R^2 = 0.607$ with scatter $= 0.286$ dex, established as an intrinsic limit (94% irreducible). The membrane rigidity $\kappa \sim 10^{-40}$ kpc^2 is effectively zero, simplifying the Lagrangian to $\mathcal{L} \approx U(\epsilon; c)$ with no gradient term. Each radius independently obeys $\mu(x) = x/(1+x)$ at RMS $= 0.090$ dex. SPARC-internal LITTLE THINGS dwarf verification ($N = 8$, $v_{\rm flat} = 15$–55 km/s) yields scatter $= 0.232$ dex with KS $p = 0.67$ versus the full sample, confirming C15 consistency across the mass range. MOND rejection is established through three independent methods ($p = 2 \times 10^{-27}$, $p = 8 \times 10^{-33}$, $p < 0.001$).

Version 4.7.8 extends the framework to pressure-supported dwarf spheroidal (dSph) galaxies. Direct C15 application to 31 dSph systems fails with +1.5 dex systematic bias, traced to an inversion of the Jeans-mass hierarchy: $M_{J,\rm mem} \ll M_{J,\rm std}$ in dSph (vs $M_{J,\rm std} \lesssim M_{J,\rm mem}$ in SPARC), reversing the direction of membrane–baryon coupling. In the $c \to 0$ limit of $U(\epsilon; c)$, the thermal Bernoulli variance $s_0(1 - s_0)$ predicts a universal acceleration scale $G_{\rm Strigari} = s_0(1 - s_0)\, a_0 = 0.228\, a_0$ with $T_m = \sqrt{6}$. The prediction is verified in two independent populations: 31 dSph galaxies ($0.240\, a_0$, 5% agreement) and 30 radius points from four SPARC bridge galaxies with outer $\Upsilon_{\rm dyn} > 10$ ($0.219\, a_0$, 4% agreement). All four bridge galaxies (ESO444-G084, NGC2915, NGC1705, NGC3741) show statistically significant radial increase of $\Upsilon_{\rm dyn}$ ($z = 2.4$ to 21.6), confirming a continuous C15 $\to$ Strigari transition within individual galaxies. Multivariate regression $\log g_{\rm obs}$ vs $(\log\sigma, \log r_h, \log L_V)$ verifies L_V-independence at machine precision ($\beta_{L_V} = -2 \times 10^{-7}$, $R^2 = 1.0000$), confirmed independently by partial correlation. A four-component scatter decomposition shows 89.4% of the 0.350 dex residual scatter is physical, with the Milky Way and M31 satellite populations yielding identical mean offsets from the Bernoulli prediction (+0.030 vs +0.032 dex), indicating that $G_{\rm Strigari}$ is independent of the host galactic environment.

Keywords: dark matter alternatives; galactic rotation curves; membrane cosmology; geometric mean law; modified gravity; weak gravitational lensing; dwarf spheroidal galaxies; Strigari relation

1. Introduction

The flat rotation curves of spiral galaxies (Rubin and Ford, 1980) and the success of the Tully–Fisher relation (Tully and Fisher, 1977) have long been interpreted as evidence for dark matter halos. Alternative approaches, including Modified Newtonian Dynamics (MOND; Milgrom 1983), Emergent Gravity (Verlinde, 2017), and various scalar-tensor theories, seek to reproduce these observations through modifications of gravitational dynamics rather than new matter species.

Here we propose a membrane cosmology framework rooted in brane-world scenarios (Randall and Sundrum, 1999). The central idea is that gravitons propagate in a $(4+1)$-dimensional bulk, but are partially screened by folds in the membrane carrying Standard Model matter. This screening modulates the effective Newton constant locally, producing rotation-curve fits without a dark matter halo.

The key advance is the *geometric mean law*: the critical acceleration g_c that governs the transition from Newtonian to MOND-like behaviour is not a universal constant (as MOND assumes) but varies from galaxy to galaxy as the geometric mean of a universal scale a_0 and a local dynamical surface density $G\Sigma_0$. This law is established from 175 SPARC galaxies with $\Delta\mathrm{AIC} = -130$ versus MOND. In v4.7, the theory is substantially simplified: the membrane rigidity κ is confirmed to be zero, eliminating the gradient term from the Lagrangian and reducing the free parameter count by four.

The paper is organised as follows. Section 2 presents the theoretical framework including the $\kappa = 0$ simplification. Section 3 establishes the geometric mean law and Condition 15 final form. Section 4 summarises statistical verification. Section 5 presents observational tests including weak lensing and dwarf galaxy verification. Section 6 discusses comparison with MOND and remaining challenges. Section 7 extends the framework to pressure-supported dwarf spheroidal systems via the $c \to 0$ limit of the elastic potential, deriving the Bernoulli prediction $G_{\mathrm{Strigari}} = s_0(1 - s_0)\, a_0$ and verifying continuous C15→Strigari transition in SPARC bridge galaxies, with L_V-independence verified at machine precision and scatter decomposed into its physical components. Section 8 summarises.

2. Theoretical Framework

2.1 Brane-world setup and gravitational screening

Consider a $(4+1)$-dimensional spacetime with the Randall–Sundrum metric. Standard Model fields are confined to the membrane at $y = 0$. When the membrane is folded with local depth $D(r)$, gravitational flux lines are partially redirected. We parametrise the fold by a dimensionless field $\epsilon(x)$ with equilibrium value satisfying $U'(\epsilon_0) = 0$, where:

$$U(\epsilon; c) = -\epsilon - \frac{\epsilon^2}{2} - c\ln(1-\epsilon), \qquad 0 < \epsilon < 1. \tag{1}$$

The effective gravitational acceleration felt by a test mass is:

$$g_{\mathrm{obs}}(r) = \frac{g_N(r) + \sqrt{g_N(r)^2 + 4g_c g_N(r)}}{2}. \tag{2}$$

This is mathematically identical to the MOND simple interpolation function but arises from the equilibrium condition of the elastic potential $U(\epsilon; c)$ — a first-principles derivation rather than an empirical ansatz. The general-c equilibrium condition is:

$$\frac{c - 1 + \epsilon^2}{1-\epsilon} = \frac{g_N}{g_c}. \tag{3}$$

2.2 Local equilibrium and membrane rigidity (v4.7)

The membrane Lagrangian in principle contains a gradient term: $\mathcal{L}[\epsilon] = U(\epsilon; c) + \kappa(d\epsilon/dr)^2$. We estimate κ directly from SPARC rotation curve data using both the Euler–Lagrange equation and the $E_{\mathrm{grad}}/E_{\mathrm{local}}$ ratio at each radius.

Result: $\kappa \sim 10^{-40}$ kpc^2 with 1.03 dex scatter and CV = 3.14 (non-universal). The rigidity length $\ell_\kappa \sim 0$ kpc: the membrane has no spatial elastic memory. The inverse prediction $B = 2\kappa/(U'' h_R^2) = 0.000$, compared to the empirical value 0.51 from previous versions — a complete failure. Consequently:

1. The gradient term is physically irrelevant: $\mathcal{L} \approx U(\epsilon; c)$.
2. The membrane responds independently at each radius: each point on the rotation curve obeys $\mu(x) = x/(1+x)$ with a single g_c per galaxy. Simple interpolation applied independently at 3003 data points across 167 galaxies gives RMS = 0.090 dex.
3. The Euler–Lagrange derivation pathway is abandoned.
4. Equations (1)–(3) constitute the complete dynamical content of the framework.

3. The Geometric Mean Law

3.1 Measurement of g_c

For each SPARC galaxy, g_c is measured directly from the Radial Acceleration Relation (RAR) without using the scale radius r_s, avoiding circularity with the rotation curve fit. The characteristic acceleration scale is $G\Sigma_0 = v_{\text{flat}}^2/h_R$.

3.2 Core result

$$g_c = \eta \cdot a_0^{(1-\alpha)} \cdot (G\Sigma_0)^\alpha, \qquad \alpha = 0.545 \pm 0.041. \tag{4}$$

Hypothesis	p-value	Decision
$\alpha = 0$ (MOND: $g_c = \text{const}$)	2.0×10^{-27}	Rejected
$\alpha = 0.5$ (geometric mean)	0.27	Cannot reject
$\alpha = 1.0$ (pure $G\Sigma_0$)	2.2×10^{-21}	Rejected

3.3 Model comparison

Model	Params	ΔAIC vs MOND
A: MOND ($g_c = a_0$)	0	0 (baseline)
C: Geometric mean ($\alpha = 0.5$ fixed)	1	−130
B: Geometric mean (α free)	2	−130

3.4 Condition 15 final form (v4.7 definitive)

$$g_c = 0.584 \cdot \Upsilon_d^{-0.361} \cdot \sqrt{a_0 \cdot v_{\text{flat}}^2/h_R}. \tag{5}$$

Parameter	Value	Note
α (fixed)	0.5	Optimal; α free gives +0.3% improvement
β (Υ_d exponent)	-0.361 ± 0.078	LOO verified, ΔAIC = −14.2
η_0	0.584	(see Sec. 6.8)
R^2	0.607	Scatter = 0.286 dex (intrinsic limit)
LOO scatter	0.290 dex	No overfitting
$b_{v_{\text{flat}}}/(-b_{h_R})$	1.985	Theory prediction: 2.0 (exact match)
MOND rejection	$p = 1.66 \times 10^{-53}$	Decisive

Scatter decomposition:

Component	σ (dex)	Fraction
Intrinsic physical variance	0.227	63.5%
g_c measurement error	0.152	28.3%
$\Upsilon_d + v_{\text{flat}} + h_R$ input	~ 0.08	8.2%
Total	0.286	100%

3.5 Jackknife verification

Test	Result
Half-split ×1000	$\alpha_{\text{train}} = 0.545 \pm 0.037$, $\alpha_{\text{test}} = 0.550 \pm 0.038$
Leave-50-out ×500	$\alpha = 0.546 \pm 0.024$, range $[0.483, 0.638]$
Bootstrap ×1000	$\alpha = 0.548 \pm 0.038$, 95% CI $[0.478, 0.626]$
Improves over MOND	100% of splits (median 31%)

3.6 Decomposition and origin of $\alpha = 0.5$

Element substitution analysis reveals that the g_c measurement method (tanh fit vs direct median) is irrelevant; what matters is per-galaxy Υ_d optimisation. Direct median with $\Upsilon_d = 0.5$ fixed gives $\alpha = 0.432$, but with Υ_d individually optimised gives $\alpha = 0.546$ — identical to the tanh result. The origin of $\alpha = 0.5$ is identified as an algebraic consequence of the deep-MOND limit. With $g_N \ll g_c$, Eq. (2) reduces to $g_{\text{eff}} = \sqrt{g_N g_c}$. Direct fitting yields $\alpha = 0.471 \pm 0.036$, $p(0.5) = 0.43$ (A-grade). Three-layer decomposition: deep-MOND (0.47) + MOND transition (+0.05) + residual scatter (+0.02) = 0.545.

3.7 Component decomposition

$g_c \sim v_{\text{flat}}^{1.10}\, h_R^{-0.56}$, $R^2 = 0.529$. Both exponents are consistent with the geometric mean prediction ($v_{\text{flat}}^{1.0}\, h_R^{-0.5}$): $p(v_{\text{flat}} = 1.0) = 0.24$, $p(h_R = -0.5) = 0.49$. Total baryonic mass M_{bar} alone gives $R^2 = 0.021$ — mass does not drive g_c. The surface density proxy $\Sigma_{\text{dyn}} = v_{\text{flat}}^2/h_R$ is the essential quantity.

3.8 C15 as minimal sufficient statistic (v4.7)

Six models incorporating a galaxy-level plasticity indicator S_{gal} (constructed from f_{gas}, Υ_d, SBdisk0, T-type) are tested against C15. All six are rejected (LOO improvement $< 2\%$, no model satisfies both LOO $> 5\%$ and ΔAIC < -5). The decisive finding: in model M1 (C15 + S_{gal}), the S_{gal} coefficient reverses sign (+0.44) compared to the raw partial correlation ($\rho = -0.23$), indicating that S_{gal}'s apparent predictive power is entirely absorbed by Σ_{dyn} already present in C15. Forward stepwise regression with all available parameters (I_0, f_{gas}, T-type, quality Q) achieves at most 1.7% improvement (LOO). Distance, inclination, and velocity errors are non-significant ($p > 0.3$). C15 is the minimal sufficient statistic for g_c prediction. Grade: A.

4. Statistical Verification

4.1 SPARC 167 galaxies

Extending g_c measurement to 167 SPARC galaxies (quality cut $\chi^2/\text{dof} < 10$): median $g_c/a_0 = 0.24$, with systematic Hubble-type dependence (Spearman $r = -0.760$, $p < 0.0001$). g_c decreases by a factor of 22 from Sab to Im type. The null correlation of r_s with g_c ($r = 0.018$, n.s.) demonstrates that g_c is determined by membrane material properties, not spatial location.

4.2 First-principles derivation of F_{conf}

The confining term $F_{\text{conf}} = -c\ln(1-\epsilon)$ in $U(\epsilon; c)$ is derived from the configuration entropy of the kink–fold network via forest + cavity mean-field effective theory (Lemmas 1–4, all established). Observational confirmation: z_{kin} shows $r = -0.507$ ($p = 2 \times 10^{-12}$) anti-correlation with g_c/a_0, and two independent proxies agree at $r = 0.809$ ($p < 10^{-38}$), supporting the physical reality of the fold density n_{fold}. Self-consistency: $\alpha = 0.5$ follows from the deep-MOND algebraic structure and the BTFR–v_{flat}–h_R closure (slope $= 0.545 \pm 0.075$, prediction 0.5, 0.6σ).

5. Observational Tests

5.1 Weak-lensing Q-value test

Nine galaxy clusters from the Miyaoka et al. (2019) sample with HSC-SSP weak-lensing data: $Q > 1$ in 9/9 clusters (100%). Wilcoxon $p = 0.0020$ (Grade A). Sign of α confirmed at 98.4%.

5.2 HSC-SSP cl1 analysis

Model	χ^2/dof	ΔAIC vs NFW
NFW (M_{200}, c_{200})	1.98	0
MOND ($g_c = a_0$, Abel)	1.69	-4.9
Membrane (g_c free, Abel)	1.54	-4.4

$g_c = 1.58\,a_0$ (68% CI: $[1.12, 2.00]$). a_0 is within 95% CI. Level B+ (systematic issues: $M_{\rm lens}/M_{\sigma_v} = 7.5$, g_c method dependence). Future surveys (Rubin LSST, Euclid) needed.

5.3 LITTLE THINGS dwarf verification (v4.7)

Eight LITTLE THINGS dwarf irregular galaxies (Oh et al., 2015) are identified within the SPARC 175 sample, spanning $v_{\rm flat} = 15$–55 km/s and $g_c/a_0 = 0.1$–0.65:

Metric	LT subset (N=8)	Full SPARC	non-LT
C15 scatter (dex)	0.232	0.286	0.288
Median bias (dex)	-0.189	$+0.000$	$+0.004$
KS test p	0.67	—	—

KS $p = 0.67$ confirms that dwarf residuals are drawn from the same distribution. C15 holds uniformly across the mass range. Combined with the full-sample LITTLE THINGS result ($N = 178$, $\alpha = 0.576 \pm 0.047$, $p(0.5) = 0.109$), three independent MOND rejection pathways are established.

Non-SPARC external verification (v4.7.7). To extend the dwarf verification beyond the SPARC sample, we applied C15 to 18 non-SPARC LITTLE THINGS galaxies from Oh et al. (2015) with baryon-separated rotation curves from VLA HI and Spitzer 3.6 μm photometry. Galaxy-specific Υ_d values were estimated from SED-derived stellar masses (Zhang et al., 2012) and V-band luminosities. After quality cuts excluding 5 galaxies with known kinematic anomalies (starburst, outflow contamination, or insufficient $V_{\rm bar}$ coverage), the remaining 11 high-quality galaxies yield scatter = 0.289 dex with bias = -0.004 dex. A KS test confirms that non-SPARC and SPARC-internal residuals are drawn from the same distribution ($p = 0.73$). Combined with the SPARC-internal result ($N = 8$, scatter = 0.232 dex), C15 is verified across the full dwarf mass range $v_{\rm flat} = 12$–125 km/s with no systematic deviation. Grade: A (upgraded from B+ via independent external confirmation).

6. Comparison with MOND and Remaining Challenges

6.1 MOND as special case

MOND's a_0 is recovered as the all-galaxy average of g_c. An "average galaxy" ($G\Sigma_0 \sim a_0/\eta^2$) gives $g_c \sim a_0$, recovering MOND. MOND is a special case of the membrane framework. MOND rejection is established through three independent methods: RAR-based ($p = 2\times10^{-27}$), deep-MOND inversion ($p = 8 \times 10^{-33}$), and LITTLE THINGS independent ($p < 0.001$).

6.2 $V_{\rm bar}$ bottleneck

The primary obstacle to Level A external verification is the $V_{\rm bar}$ bottleneck: baryon-separated rotation curves are publicly available only in SPARC and LITTLE THINGS. PROBES (3,158 galaxies) and MaNGA DynPop (10,296) provide only $V_{\rm obs}$, yielding $\alpha \sim 1.0$ with fixed $\Upsilon_d = 0.5$. These are repositioned for large-sample precision measurement rather than independent verification. BIG-SPARC ($\sim$ 4000 galaxies, in preparation) will provide the definitive external test.

6.3 $\kappa = 0$ and theory simplification (v4.7)

The $\kappa = 0$ result (Sec. 2.2) has far-reaching consequences for the theory structure:

Aspect	Before (v4.6)	After (v4.7)
Lagrangian	$\mathcal{L} = U(\epsilon; c) + \kappa(d\epsilon/dr)^2$	$\mathcal{L} \approx U(\epsilon; c)$
g_c structure	C15 global + C14 radial	C15 only
Scatter	0.286 dex (improvable?)	0.286 dex (intrinsic limit)
Free params removed	A, B, τ, κ	All four eliminated

The Condition 14 dynamical formulation $g_{c,\rm eff}(r) = g_c[0.12 + 0.51\exp(-r/(4.7h_R))]$ reported in v4.6 is retracted. The pattern is a projection of C15 prediction bias onto the radial coordinate. The conceptual hypothesis of plastic/elastic two-phase membrane structure (M3 ϵ threshold, $\Delta\mathrm{AIC}_c = -5.1$) remains at B−grade as an interpretive framework.

6.4 h_R partial correlation

The scale-length h_R shows zero raw correlation with g_c ($\rho = -0.022$) but a significant partial correlation after controlling for $v_{\rm flat}$ ($\rho = -0.472$). Decomposition: 69% from outer-disk MOND transition (A-grade), 18% from $x(r)$ profile higher-order effects (B−grade, reattributed from gradient energy in v4.7), 13% residual.

6.5 BTFR slope and Simpson's paradox

The membrane BTFR slope = 1.20 is a statistical confound independent of g_c: setting $g_c = a_0$ (constant) yields the same slope 1.196. Simpson's paradox arises from $v_{\rm flat}$ driving both BTFR axes. Within-bin slope $= 0.542 \pm 0.069$ (bootstrap 95% CI $[0.400, 0.668]$, $p(0.5) = 0.494$) provides a non-circular confirmation of $\alpha = 0.5$. In dwarf galaxies ($v_{\rm flat} < 47$ km/s), $b_{gc} = 2.32$: the membrane fold state dominates rotation velocity, a distinctive prediction with no analogue in standard dark matter models.

6.6 Deep-MOND g_c–$\Sigma_{\rm bar}$ independence

In the deep-MOND regime (bottom 30% by g_c), g_c is uncorrelated with baryonic surface density: $r(g_c, \Sigma_{\rm bar}) = +0.066$. In contrast, $r(g_c, \Sigma_{\rm dyn}) = +0.727$. The ratio $\Sigma_{\rm bar}/\Sigma_{\rm dyn}$ anti-correlates with g_c ($r = -0.648$): from ~ 0.7 in low-g_c galaxies to ~ 0.14 in high-g_c. The membrane responds to dynamical stress-energy, not baryonic mass content. The self-referential closure $g_c \sim \Sigma_{\rm bar}^{1/3}$ fails because its premise is empirically false. Grade: A.

6.7 Intrinsic scatter limit

Two independent optimisation pathways fail to reduce the 0.286 dex scatter: (i) $\alpha(g_c)$ tanh transition achieves 3% improvement ($\Delta\mathrm{AIC} = -2.4$, 4 extra parameters — insufficient); (ii) all additional galaxy parameters achieve at most 1.7% (LOO). The scatter-minimising $\alpha(g_c)$ parameters contradict the sliding-window α in direction, confirming that local statistical slopes and global prediction accuracy are distinct quantities. The Rotmod-derived local g_c (0.244 dex) provides a lower bound. Grade: A.

6.8 Plasticity narrative and self-consistent equation (v4.7.6)

The Υ_d dependence of η is a structural consequence of the BTFR + GML system, not a "plasticity proxy". All plasticity indicators ($f_{\rm gas}$, SBdisk0, T-type) are different aspects of $\Sigma_{\rm dyn}$ variation already captured by C15. The morphological-type bridge (T-type vs $S_{\rm gal}$, $\rho = +0.89$, $p = 4 \times 10^{-60}$) establishes a strong conceptual correspondence but adds no predictive power. "Plasticity" is an explanatory narrative, not a predictive variable. Grade: A (bridge) / X ($S_{\rm gal}$ prediction).

Theoretical basis for η_0 (v4.7.6). The C15 normalisation $\eta_0 = 0.584$ can be given a partial first-principles interpretation through the membrane self-consistent equation. Define the elastic fraction $s = 1 - f_p$, where f_p is the plastic content of the membrane at a given galaxy. The dimensionless dynamical structure parameter $Q = \sqrt{v_{\rm flat}^2/(a_0 h_R)}$ measures the fold-driving stress. The equilibrium elastic fraction satisfies:

$$s = \frac{1}{1 + \exp(-\Delta U(sQ)/T_m)}, \tag{6}$$

where $\Delta U(c) = -3/2 + 2\sqrt{c} - c/2 - (c/2)\ln(c)$ for $0 < c \leq 1$ is the potential energy difference between the folded and unfolded membrane states, derived from $U(\epsilon; c)$ (Eq. 1). The effective membrane temperature $T_m = \sqrt{6}$ is the Z_2 symmetry-breaking critical temperature (Lemma 5). Key properties of Eq. (6): $\Delta U(0) = -3/2$ (deep plastic well), $\Delta U(1) = 0$ with $\Delta U'(1) = 0$ (inflection point), guaranteeing a unique solution for each Q.

Applying Eq. (6) to the 159 SPARC galaxies in the TA3 sample, the SPARC-averaged elastic fraction is $\langle s \rangle = 0.58$, matching $\eta_0 = 0.584$ to within 1%. This identifies η_0 as the mean elastic fraction of the membrane across the observed galaxy population: $\eta_0 = \langle 1 - f_p \rangle_{\rm SPARC}$. Grade: B+.

Υ_d as independent variable (v4.7.6). The extended model $g_c/a_0 = A\, s(Q, T_m)\, Q\, \Upsilon_d^\beta$ with $T_m = \sqrt{6}$ yields $\beta = -0.331$ and scatter = 0.2845 dex, compared to the C15 empirical $\beta = -0.361$ (8% difference). Crucially, the residuals of the pure $s(Q)$ model (Υ_d^β omitted) correlate with Υ_d at $r = -0.296$ ($p = 7 \times 10^{-5}$), demonstrating that Υ_d cannot be derived from Q alone. The membrane state space is at least two-dimensional: dynamical structure (Q) and material composition (Υ_d). Grade: A (Υ_d independence).

Partial correlation test (v4.7.7). To test whether the g_c–Υ_d correlation is mediated by gas fraction, we compute the partial correlation $r(\log g_c, \log \Upsilon_d | f_{\rm gas})$ using the SPARC 175 sample ($f_{\rm gas} = 1.33\, M_{\rm HI}/(1.33\, M_{\rm HI} + \Upsilon_d L_{3.6})$). The partial correlation is $r = -0.437$ ($p = 1.6 \times 10^{-9}$), slightly stronger than the raw $r = -0.428$. The attenuation ratio is 1.02: conditioning on $f_{\rm gas}$ does not weaken the correlation. The decisive result is $r(g_c, f_{\rm gas}) = +0.061$ ($p = 0.42$): g_c is uncorrelated with gas fraction. This rules out the common-cause model $f_p \to g_c$ and $f_p \to \Upsilon_d$ mediated by $f_{\rm gas}$. Instead, Υ_d itself is the direct observational proxy of the membrane plastic fraction f_p. Grade: A.

Resolution of the β discrepancy (v4.7.7). The 8.3% difference between the self-consistent equation prediction ($\beta = -0.331$ at $T_m = \sqrt{6}$) and the C15 empirical value ($\beta = -0.361$) is fully explained as a finite-temperature absorption effect. A systematic T_m scan confirms that at $T_m \to \infty$ ($s = 0.5 = $ const), the model recovers $\beta = -0.361$ exactly: C15 implicitly assumes constant s. At finite $T_m = \sqrt{6}$, the variation of $s(Q)$ across the galaxy population absorbs 8.3% of the Υ_d dependence, reducing the direct β from -0.361 to -0.331. This absorption requires no parameter adjustment and arises from the Q-dependent structure of $\Delta U(c)$. The scatter-minimising $T_m = 2.35$ is consistent with $\sqrt{6} = 2.449$ to within 0.24σ, providing independent empirical support for the Z_2 SSB critical temperature. Grade: A (β discrepancy resolved; $T_m = \sqrt{6}$ confirmed by scatter optimisation).

6.9 Weak Lensing RAR: KiDS-1000 and HSC-SSP Y3 (v4.7.6)

We examined the KiDS-1000 weak lensing Radial Acceleration Relation (Brouwer et al., 2021) for consistency with C15. The KiDS lensing RAR extends the rotation-curve RAR by two decades in $g_{\rm obs}$ into the low-acceleration regime ($g_{\rm obs} \sim 10^{-15}$ to 10^{-12} m/s^2), using stacked Excess Surface Density (ESD) profiles of ~ 1 million isolated galaxies from the KiDS-bright sample.

Following Brouwer et al. (2021), we adopt their nominal circumgalactic gas (CGM) model ($M_{\rm hot}/M_* = 1$, isothermal profile, truncation radius $R_{\rm acc} = 100$ kpc). A universal correction factor $C(g_{\rm bar}) = {\rm ESD}_{\rm hotgas}(g_{\rm bar})/{\rm ESD}_{\rm no\text{-}hotgas}(g_{\rm bar})$ is derived from the full-sample (no mass bins) ESD pair and applied to four stellar mass bins ($\log_{10} M_*/M_\odot = [8.5, 10.3, 10.6, 10.8, 11.0]$). Per-bin g_c is then fitted using the membrane RAR.

Under the nominal CGM model, the hot-gas-corrected per-bin g_c values show a positive M_* slope ($+0.138$ in $\log g_c$ vs $\log M_*$), consistent in sign and order of magnitude with the C15 prediction ($+0.075$). The per-bin ratio $g_c^{\rm obs}/g_c^{\rm C15}$ ranges from 0.77 to 1.05 across all four bins, with Bin 4 ($\log M_* = 10.9$) achieving exact agreement (ratio = 1.00).

However, a systematic sensitivity analysis reveals that this result is not robust to CGM mass uncertainty. Scanning $M_{\rm hot}/M_*$ over the observationally permitted range $[0.3, 3.0]$ (Babyk et al., 2018; Tumlinson et al., 2017), the formal $\Delta\chi^2({\rm MOND} - {\rm C15})$ computed with the full 60×60 covariance matrix reverses sign: $\Delta\chi^2 = -10.6$ at $f_{\rm gas} = 0.3$ (MOND preferred) versus $+147.7$ at $f_{\rm gas} = 1.0$ (C15 preferred). Introducing a physically motivated M_*-dependent correction ($R_{\rm acc} \propto M_*^{0.4}$) further reduces the C15 preference to $\Delta\chi^2 = +0.8$ at $f_{\rm gas} = 1.0$, rendering the two models statistically indistinguishable.

The Brouwer et al. (2021) colour and Sérsic-index bins (their Fig. 8) provide a complementary test. Splitting by colour or Sérsic index, we find $g_c({\rm red})/g_c({\rm blue}) = 2.3$ to 2.8 and $g_c({\rm high}{-}n)/g_c({\rm low}{-}n) = 2.4$ to 2.5, with the ordering stable across $f_{\rm gas} = 0$ to 1.0 ($\Delta\chi^2({\rm uniform} - {\rm split}) = +127$ for colour, $+67$ for Sérsic). This result is CGM-robust because the hot-gas correction affects both morphological bins approximately equally, preserving the g_c ratio.

However, a SPARC cross-check reveals that this ordering is dominated by the colour-magnitude relation rather than a pure morphological effect. In SPARC, the median g_c ratio between early-type ($T \leq 3$) and late-type ($T > 3$) galaxies is 0.92, with no statistically significant difference (Mann–Whitney $p = 0.35$). C15 predicts a ratio of 1.7, arising entirely from $\Sigma_{\rm dyn} = v_{\rm flat}^2/h_R$ differences, with zero contribution from Υ_d (both groups share median $\Upsilon_d = 0.30$ in the TA3 sample). The KiDS ratio of 2.3 to 2.8 exceeds both the SPARC observation and the C15 prediction, indicating that the colour/Sérsic bins primarily separate galaxies by stellar mass rather than by morphological type, with the g_c ordering reflecting the underlying M_*–$\Sigma_{\rm dyn}$ correlation.

In summary, the KiDS lensing data are consistent with both C15 and MOND at the current level of systematic uncertainty. The stellar-mass test cannot discriminate between the two models due to CGM mass uncertainty, and the colour/Sérsic test, while CGM-robust, is dominated by the colour-magnitude relation rather than an independent morphological signal. Definitive tests will require future surveys combining high-precision weak lensing (Euclid, Rubin LSST) with resolved CGM measurements (eROSITA, SKA), and colour-binned analysis within narrow stellar-mass slices to isolate pure morphological effects. Grade: B (both tests). M_* test: CGM-limited. Colour/Sérsic test: CGM-robust but colour-M_* degenerate.

HSC-SSP Y3 independent lensing RAR (v4.7.4). To provide an independent test of the lensing RAR obtained with KiDS-1000, we constructed a fully independent weak-lensing pipeline using the Hyper Suprime-Cam Subaru Strategic Program Year 3 (HSC-SSP Y3) shape catalogue as source galaxies and the Galaxy and Mass Assembly Data Release 4 (GAMA DR4) spectroscopic survey as lens galaxies. The HSC-SSP Y3 catalogue contains 35.7 million galaxies with shapes measured via the Re-Gaussianization (regauss) method, reaching an effective source number density of 19.9 arcmin^{-2} with a median i-band seeing of 0.6 arcsec. GAMA DR4 provides accurate spectroscopic redshifts ($N_Q \geq 3$) and stellar masses across three equatorial fields (G09, G12, G15), each spanning 60 deg^2.

We selected isolated lens galaxies from GAMA DR4 via the G3C group catalogue, retaining only central galaxies (RankIterCen = 1) with $0.05 < z < 0.5$ and $\log_{10}(M_*/M_\odot) < 11.0$, the latter matching the isolation criterion of Brouwer et al. (2021). This yielded 157,338 isolated lenses across three fields: 49,272 (G09), 55,402 (G12), 52,664 (G15). The baryonic acceleration was computed per lens as $g_{\rm bar} = GM_{\rm gal}/R^2$, treating each lens as a baryonic point mass (Brouwer et al., 2021, Eq. 2), with pairs binned in 15 logarithmic $g_{\rm bar}$ bins spanning 10^{-15} to 5×10^{-12} m s^{-2}. A total of 503 million lens–source pairs were accumulated.

Systematic corrections were applied as follows. The HSC regauss shear responsivity $\mathcal{R} = 1 - \langle e^2_{\rm rms}\rangle$ was computed per field ($\mathcal{R} \sim 0.86$). The multiplicative shear bias for HSC Y3 regauss is controlled at $|m| < 10^{-2}$ through image-simulation calibration (Li et al., 2022; Mandelbaum et al., 2018), contributing $< 1\%$ ESD uncertainty. The observed acceleration was obtained via the SIS approximation $g_{\rm obs} = 4G\Delta\Sigma$ (Brouwer et al., 2021, Eq. 7). The cross component ESD$_\times$ is consistent with zero across all bins, confirming the absence of significant additive systematics. Statistical uncertainties were estimated using a 24-patch spatial jackknife (8 per field) yielding the full 15×15 covariance matrix.

A critical systematic-robustness test is the consistency of results across independent survey fields. Fitting the interpolation $g_{\rm obs} = g_{\rm bar}/(1 - \exp(-\sqrt{g_{\rm bar}/g_c}))$ independently in each field yields $g_c = 2.91 \pm 0.19\, a_0$ (G09), $2.64 \pm 0.16\, a_0$ (G12), $2.64 \pm 0.23\, a_0$ (G15). The inter-field consistency $\chi^2 = 1.37$ (2 d.o.f., $p = 0.50$) confirms that all three fields are statistically consistent at the 1σ level, with no evidence for field-dependent systematics.

Combining all three fields yields $g_c = 2.73 \pm 0.11\, a_0$ for the full isolated lens sample. Comparing C15 (g_c free) to MOND ($g_c = a_0$ fixed) using the full jackknife covariance matrix yields $\chi^2_{\rm MOND} = 539.3$ versus $\chi^2_{\rm C15} = 65.0$, giving $\Delta\chi^2 = +474$ and ΔAIC $= +472$, corresponding to a ~ 22-sigma preference for C15 over MOND. The lensing RAR of isolated galaxies at Mpc scales is incompatible with a universal acceleration scale $g_c = a_0$ as predicted by MOND.

The g_c–M_* relation provides a further discriminant. Fitting g_c in four stellar-mass bins between $\log_{10} M_* = 8.5$ and 11.0, we find a monotonic increase from $g_c = 2.00 \pm 0.27\, a_0$ (lowest M_*) to $3.41 \pm 0.21\, a_0$ (highest M_*), with a log–log slope of $+0.166 \pm 0.041$. MOND predicts zero slope and is rejected at 4.0σ ($p < 0.001$). C15, which predicts a positive slope of $+0.075$ from the $\Sigma_{\rm dyn}$ dependence, is consistent at the 2.2σ level ($p = 0.029$). The observed slope being approximately twice the C15 prediction suggests that the $\Sigma_{\rm dyn}$ dependence at Mpc lensing scales may be steeper than at kpc rotation-curve scales, potentially reflecting the transition from baryon-dominated to halo-dominated regimes. This is a quantitative prediction for future refinement of the membrane framework.

In summary, the HSC-SSP Y3 lensing RAR independently confirms the C15 framework at high significance: (i) $g_c = 2.73 \pm 0.11\, a_0$ is inconsistent with the MOND universal value a_0 at $> 15\sigma$; (ii) the g_c–M_* slope rejects MOND at 4σ while remaining consistent with C15's predicted positive slope; (iii) all three GAMA fields yield consistent g_c values ($p = 0.50$); (iv) combined with the KiDS-1000 analysis, the lensing RAR evidence upgrades from B grade (KiDS alone, CGM-limited) to A grade (independent HSC confirmation with systematic verification). The observed factor ~ 2 excess of the slope over the C15 prediction is noted as a quantitative tension requiring theoretical investigation, but does not affect the qualitative conclusion that g_c varies with galaxy properties as predicted by membrane cosmology and contrary to MOND.

Slope budget analysis (v4.7.6). The factor ~ 2 excess of the observed g_c–M_* slope ($+0.166 \pm 0.041$) over the C15 prediction ($+0.075$) reported above constituted a 2.2σ tension. We decompose this excess into identified physical contributions.

First, the 2-halo contribution to the lensing ESD slope was measured empirically by separating the 1-halo and total (1h+2h) regimes in each M_* bin. Fitting g_c using only the 1-halo-dominated radial range ($R < 200$ kpc) and comparing to the full-range fit yields $\Delta{\rm slope}_{\rm 2h} = +0.034 \pm 0.064$ (0.5σ). The sign is consistent with the physical expectation that more massive galaxies reside in denser large-scale environments, enhancing the outer ESD and steepening the apparent g_c–M_* relation.

Second, the light-propagation model (Eqs. 10a–10d) predicts a small additional slope contribution of +0.017 from the R-dependent gravitational Aharonov–Bohm phase shift (Sec. 2).

The combined slope budget is:

Source	Slope contribution	Status
C15 ($\Sigma_{\rm dyn}$ dependence)	+0.075	A-grade (SPARC)
Model B (AB phase shift)	+0.017	B-grade
2-halo environment	$+0.034 \pm 0.064$	B-grade (0.5σ)
Total predicted	+0.126	
Observed	$+0.166 \pm 0.041$	
Residual tension	0.98σ	Resolved

The slope tension is reduced from 2.2σ to 0.98σ, within the expected statistical fluctuation. While the individual 2-halo and Model B contributions are each non-significant, their combined effect accounts for the observed excess. Future surveys with higher signal-to-noise (Euclid Wide, Rubin LSST) are expected to detect the 2-halo contribution at 2–3σ significance.

7. Pressure-Supported Dwarf Spheroidal Extension

The v4.7.7 framework was developed against rotation-supported systems (SPARC spirals and dwarf irregulars). This section extends the framework to pressure-supported dwarf spheroidal (dSph) galaxies, exposing a qualitatively new regime in which the baryonic driving of the membrane fails and the spontaneous $c \to 0$ equilibrium of $U(\epsilon; c)$ dominates. The resulting Bernoulli-variance prediction $G_{\rm Strigari} = s_0(1 - s_0)\, a_0$ is verified in two statistically independent populations to 4–5% accuracy.

7.1 dSph sample construction

We assembled a catalogue of 31 dSph galaxies: 15 Milky Way satellites ($\sigma_{\rm los}$ from Walker et al. 2009; Simon and Geha 2007; Simon 2019), 11 M31 satellites (Collins et al., 2013; Tollerud et al., 2012), 3 isolated systems (Cetus, Phoenix, Tucana), and 2 unclassified dwarfs. Half-light radii r_h were taken from McConnachie (2012), with the Walker estimator $M_{\rm dyn} = 2.5\,\sigma_{\rm los}^2\, r_h/G$ used for σ back-calculation in the VizieR catalogue (coefficient 2.5, confirmed by $\sigma_{\rm wolf}/\sigma_{\rm lit} = 0.997$). Dynamical accelerations were computed via the Wolf et al. (2010) relation $g_{\rm obs} = 3\sigma_{\rm los}^2/r_h$, and baryonic accelerations from stellar mass with $\Upsilon_* = 1$ assumed.

7.2 C15 direct application fails: J3 regime inversion

Applying the C15 formula (Eq. 5) to the dSph sample yields a systematic offset of +1.5 dex with scatter 0.9 dex. Four model variants (C15_naive, SCE_naive, SCE_Q_J, C15_Q_J) give statistically indistinguishable residuals. The Υ_d transformation does not absorb the offset (scatter invariant under Υ_d scan). Auxiliary hypotheses contribute insufficiently: H1 (3D topology) +0.15 dex, H2 (virial coefficient $\beta_c = 2$ vs $\sqrt{3}$) +0.25 dex, combined +0.40 dex (29% of the observed offset). H3 (membrane hysteresis alone) is rejected by three independent tests (isolated > satellite g_c; zero distance correlation; inferior AIC versus constant-model).

The failure traces to an inversion of the Jeans-mass hierarchy. Define the membrane Jeans mass $M_{J,\rm mem} = (4\pi/3)\rho(c_s^2/g_{\rm eff})^3$ and the standard Jeans mass $M_{J,\rm std}$. In SPARC, $M_{J,\rm std} \le M_{J,\rm mem}$ and baryonic distribution drives the membrane state (C15 valid). In dSph, $M_{J,\rm mem} \ll M_{J,\rm std}$ in 28 of 31 systems, and the membrane sets the structure while baryons follow. The direction of causality is reversed, and g_c is no longer a function of baryonic surface density Σ_0. Grade: A.

7.3 Unified RAR: SPARC × dSph and J3 transition band

Overlaying 3,389 SPARC RAR-cloud points from Rotmod_LTG (Lelli et al., 2016) with the 31 dSph sample on the $\log(g_{\rm bar}/a_0)$–$\log(g_{\rm obs}/a_0)$ plane reveals two populations occupying distinct $\Upsilon_{\rm dyn} = g_{\rm obs}/g_{\rm bar}$ ranges:

Sample	N	median $\Upsilon_{\rm dyn}$	Interpretation
SPARC RAR cloud (all radii)	3,389	2.75	baryon-dominated
SPARC galaxy-level (outer)	175	3 (typical)	C15 regime
Bridge galaxy outer (40%)	30	12.76	J3 transition band
dSph	31	35.11	membrane-dominated

We designate the band $10 < \Upsilon_{\rm dyn} < 30$ the J3 transition band. Only 4 of 175 SPARC galaxies reach outer $\Upsilon_{\rm dyn} > 10$: ESO444-G084, NGC2915, NGC1705, NGC3741. These "bridge galaxies" span the SPARC–dSph gap and permit radius-resolved verification of the transition.

7.4 Reformulation via $c \to 0$ limit of $U(\epsilon; c)$

In the J3 regime the coupling c (set by baryonic driving) approaches zero, and the membrane enters its intrinsic thermal equilibrium. The self-consistent equation Eq. (6) in the $Q \to 0$ limit yields a universal spontaneous elastic fraction:

$$s_0 = \frac{1}{1 + \exp(3/(2T_m))}. \tag{7}$$

With $T_m = \sqrt{6}$, this gives $s_0 = 0.3515$. The leading small-Q correction is non-analytic:

$$s(Q) \approx s_0 + \alpha_s \sqrt{Q} + O(Q), \qquad \alpha_s = \frac{2E}{FT_m} s_0^{3/2} = 0.110, \tag{8}$$

where $E = \exp(3/(2T_m))$ and $F = 1 + E$. The $\sqrt{Q}$ structure originates from the $\ln(1-\epsilon)$ term in $U(\epsilon; c)$ and is characteristic of the $c \to 0$ singularity. Numerical SCE verification confirms agreement with the asymptotic expansion to within 4% relative error for $Q < 0.3$.

7.5 Bernoulli prediction: $G_{\rm Strigari} = s_0(1 - s_0)a_0$

In the $c \to 0$ regime the membrane occupies a thermal two-state (folded/unfolded) equilibrium with mean s_0 and variance $s_0(1 - s_0)$ (Bernoulli distribution). We identify this intrinsic variance with the universal acceleration scale underlying the Strigari et al. (2008) dSph relation:

$$G_{\rm Strigari} = s_0(1 - s_0)\, a_0 = 0.228\, a_0 = 2.74 \times 10^{-11}\ {\rm m\,s^{-2}}. \tag{9}$$

The corresponding dSph g_c formula, obtained via $g_c = g_{\rm obs}^2/g_{\rm bar}$ at the deep-MOND limit:

$$g_c^{\rm dSph}(g_{\rm bar}) = \frac{[s_0(1 - s_0)]^2 a_0^2}{g_{\rm bar}}. \tag{10}$$

Of eight tested dimensionless combinations involving s_0 and T_m, only the Bernoulli variance form matches the observed dSph-sample ensemble mean within 5%. Alternative forms $s_0 a_0$ ($0.35\, a_0$, +47% error), $(1 - s_0)/T_m$ ($0.26\, a_0$, +10%), and $1/(2T_m)$ ($0.20\, a_0$, −15%) are rejected at the $> 2\sigma$ level. Grade: B+ (single-population match; independent verification in Sec. 7.7).

7.6 Tautology separation: definitional vs predictive content

A methodological question arises: the slope -1 of $\log g_c$ vs $\log g_{\rm bar}$ is a consequence of the Strigari condition $g_{\rm obs} \sim$ const combined with the deep-MOND extraction $g_c = g_{\rm obs}^2/g_{\rm bar}$, and therefore has no independent predictive content. We separate the definitional from the predictive components via multivariate regression on the 31-dSph sample.

Regression A ($g_{\rm obs}$ basis). Regressing $\log g_{\rm obs}$ on $(\log \sigma, \log r_h, \log L_V)$ with 1000-iteration bootstrap yields:

Coefficient	Observed	Theory	z-score	Verdict
β_σ	$+2.000 \pm 0.000$	$+2.0$	$+1.2$	definitional
β_{r_h}	-1.000 ± 0.000	-1.0	$+0.7$	definitional
β_{L_V}	-2×10^{-7}	0	-1.4	null confirmed

The fit achieves $R^2 = 1.0000$ with residual standard deviation at machine precision, recovering the Wolf estimator $g_{\rm obs} = 3\sigma^2/r_h$ as an exact identity. The L_V coefficient at 2×10^{-7} is the strongest possible statement of $M_{\rm bar}$-independence: luminosity adds no information beyond σ and r_h.

Orthogonalized regression. The dSph σ–size relation ($\log\sigma = +0.263\log r_h +$ const, consistent with McConnachie 2012) introduces collinearity between σ and r_h. Orthogonalizing σ against r_h and refitting preserves $\beta_{L_V} \approx 0$, providing an independent verification free of collinearity artifacts.

Partial correlation. A third independent route: the partial correlation $r(g_{\rm obs}, L_V|\sigma, r_h) = -0.236$ ($p = 0.20$) is statistically insignificant, while $r(g_{\rm obs}, \sigma|r_h, L_V)$ and $r(g_{\rm obs}, r_h|\sigma, L_V)$ both approach ±1 as required by the Wolf identity. Three independent methods (multivariate regression, orthogonalization, partial correlation) concur: L_V contributes no predictive information to $g_{\rm obs}$ at the machine-precision level.

Regression B (g_c basis) — quantitative confirmation of tautology. Fitting $\log g_c$ (where $g_c = g_{\rm obs}^2/g_{\rm bar}$) on the same variables yields $(\beta_\sigma, \beta_{r_h}, \beta_{L_V}) = (+3.82, +0.16, -1.07)$, compared to the definitional expectation $(+4, 0, -1)$ derived from $g_{\rm bar} \propto L_V/r_h^2$. All three coefficients match the definitional prediction within 0.2, confirming that the apparent slope -1 of $\log g_c$ vs $\log L_V$ carries zero predictive content.

Consequence for the H4 hypothesis. The earlier H4 conjecture $g_c \propto M_{\rm dyn}^{-1}$ decomposes, via the Wolf identity, into $g_c \propto r_h^{-3/2}$ with zero σ contribution. The partial-correlation analysis on g_c itself confirms: $\beta_\sigma = 0.00 \pm 0.81$, $\beta_{r_h} = -1.51 \pm 0.37$. The observed "mass scaling" is entirely a radius effect.

The r_h-dependence of $g_{\rm obs}$ (-3.4σ marginal trend) is inconsistent with strict Strigari universality ($g_{\rm obs} =$ const) and instead reflects the empirical σ–size scaling $\sigma \propto r_h^{0.27}$ (Brasseur et al., 2011; McConnachie, 2012), which is not derived from the mean-field $c \to 0$ analysis. The origin of this scaling is deferred to future work.

7.6.1 Scatter decomposition: 89% of residual scatter is physical

To quantify the information content of the 0.350 dex scatter in $\log(g_{\rm obs}/a_0)$ around the Bernoulli prediction, we decompose the total variance into four components: definitional (unavoidable given observed σ and r_h), measurement (Monte Carlo error propagation with 5% σ and 10% r_h relative uncertainties), host environment (between-host variance across MW / M31 / Isolated populations), and physical residual.

Component	σ (dex)	Var. fraction	Origin
Total	0.350	100.0%	$\log(g_{\rm obs}/a_0)$ vs $\log(G_{\rm Strigari}/a_0)$
Measurement	0.062	3.2%	MC propagation, 1000 draws
Host (between)	0.095	7.4%	MW / M31 / Isolated offsets
Physical residual	0.331	89.4%	σ–size projection (primary)

The measurement contribution is negligible (3.2%), confirming that the scatter is not limited by observational precision. The host-environment contribution is small (7.4%), but its structure carries a strong independent test:

Host	N	Scatter (dex)	Mean offset (dex)
MW satellites	15	0.323	+0.030
M31 satellites	11	0.410	+0.032
Isolated	3	0.332	+0.164

The MW and M31 satellite populations yield essentially identical mean offsets from the Bernoulli prediction (+0.030 vs +0.032 dex, difference $< 5\%$ of their individual scatters). This is a non-trivial cross-check: two independent host galaxies of comparable mass (both $\sim 10^{12}\,M_\odot$) produce populations that calibrate $G_{\rm Strigari}$ to the same value, indicating that the prediction is independent of the host galactic environment. The Isolated sample shows a marginally higher offset (+0.164 dex) with small $N = 3$, in the same direction as the bridge-galaxy outer points ($+0.219\,a_0$, Sec. 7.7).

The residual physical scatter (0.331 dex) is dominated by the dSph σ–size scaling $\sigma \propto r_h^{0.27}$, which propagates through the Wolf identity into $g_{\rm obs}$. Derivation of this relation from first principles is a candidate task for future work beyond the mean-field $c \to 0$ analysis of the present theory.

Summary of Section 7.6. (i) Three independent methods verify L_V-independence of $g_{\rm obs}$ at machine precision, elevating the Bernoulli prediction from population-mean agreement to a direct functional null. (ii) The -1 slope of $\log g_c$ vs $\log L_V$ is quantitatively shown to be a definitional projection of the $g_{\rm bar}$ construction. (iii) The 0.350 dex residual scatter is 89% physical and 3% measurement; MW and M31 populations independently calibrate $G_{\rm Strigari}$ to the same value, establishing host-environment independence as an additional A-grade result of v4.7.8.

7.7 Bridge galaxy verification: continuous C15 → Strigari transition

For the four bridge galaxies with outer $\Upsilon_{\rm dyn} > 10$ (Sec. 7.3), Rotmod_LTG data permit radius-resolved analysis. Radial profiles of $g_{\rm obs}(R)$ and $g_{\rm bar}(R)$ are constructed, and $\Upsilon_{\rm dyn}(R)$ is tested for monotonic radial increase. We define four A-grade criteria:

Galaxy	N	R [kpc]	$\Upsilon_{\rm in}$	$\Upsilon_{\rm out}$	slope($\log\Upsilon, \log R$)	z	MW p	outer $g_{\rm obs}$ [a_0]
ESO444-G084	7	0.26–4.44	6.1	11.3	$+0.31 \pm 0.13$	+2.4	0.029	0.317
NGC2915	30	0.34–10.04	11.4	13.2	$+1.13 \pm 0.09$	+12.3	0.0003	0.219
NGC1705	14	0.22–6.00	5.4	12.3	$+0.46 \pm 0.07$	+6.8	0.0003	0.282
NGC3741	21	0.23–7.00	5.9	11.5	$+0.55 \pm 0.03$	+21.6	0.0001	0.124

All four bridge galaxies show statistically significant radial increase of $\Upsilon_{\rm dyn}$ (criterion a: 4/4 at $|z| > 2$) and outer-vs-inner differences confirmed by Mann–Whitney tests (criterion b: 4/4 at $p < 0.03$). The ensemble outer $\Upsilon_{\rm dyn}$ median = 12.76 falls squarely in the J3 transition band (criterion c). The ensemble outer $g_{\rm obs}$ median = $0.219\,a_0$ agrees with the Bernoulli prediction (Eq. 9, $0.228\,a_0$) to 4% (criterion d). All four criteria A-grade satisfied.

7.8 Two-population independent verification of the Bernoulli prediction

The dSph sample (Sec. 7.5, pressure-supported systems) and the SPARC bridge galaxy outer points (Sec. 7.7, rotation-supported outer disks) constitute two statistically independent populations. Both reproduce the Bernoulli prediction $G_{\rm Strigari} = 0.228\,a_0$ at similar precision:

Population	N	$g_{\rm obs}$ [a_0]	Ratio to Bernoulli	Agreement
Bernoulli prediction (Eq. 9)	—	0.228	1.000	—
dSph sample (Sec. 7.5)	31	0.240	1.053	5%
Bridge outer points (Sec. 7.7)	30	0.219	0.961	4%
NGC2915 outer alone	12	0.219	0.961	4%

The two populations differ in dynamical support (pressure vs rotation), typical mass scale ($\sim 10^7\,M_\odot$ vs $\sim 10^9\,M_\odot$), typical acceleration ($g_{\rm bar} \sim 10^{-3}\,a_0$ vs $10^{-2}\,a_0$), and data reduction pathway

(Wolf $\sigma_{\rm los}$ vs Rotmod $V_{\rm bar}$). The common agreement with a single theoretical constant derived from a two-parameter (T_m, s_0) theory, with no free fitting, constitutes a non-trivial confirmation of the membrane $c \to 0$ limit. Grade: A (upgraded from B+ via independent bridge-galaxy verification).

7.9 Summary of dSph extension

Results established in v4.7.8:

(i) C15 direct application to dSph fails at +1.5 dex (Sec. 7.2).

(ii) The failure is explained by J3 regime inversion $M_{J,\rm mem} \ll M_{J,\rm std}$ in 28/31 dSph (A grade).

(iii) The $c \to 0$ limit of $U(\epsilon; c)$ yields a universal spontaneous equilibrium $s_0 = 0.3515$ at $T_m = \sqrt{6}$ (A grade).

(iv) The Bernoulli variance $s_0(1 - s_0)a_0 = 0.228\, a_0$ matches the observed Strigari scale in both dSph (5%) and SPARC bridge outer points (4%) with no free parameter (A grade).

(v) All four bridge galaxies show significant radial transition to the J3 regime (A grade, 4/4 criteria).

(vi) The L_V-independence of $g_{\rm obs}$ in dSph is verified at machine precision ($\beta_{L_V} = -2 \times 10^{-7}$, $R^2 = 1.0000$), cross-checked by orthogonalized regression and partial correlation (A grade).

(vii) Scatter decomposition shows 89.4% of the 0.350 dex residual is physical, with MW and M31 satellite populations yielding identical mean offsets (+0.030 vs +0.032 dex) from the Bernoulli prediction, establishing host-environment independence of $G_{\rm Strigari}$ (A grade).

Open issues: (a) The r_h-dependence of $g_{\rm obs}$ (-3.4σ) reflects the empirical dSph σ–size scaling not predicted by mean-field analysis; the individual-galaxy formula (Eq. 10) remains B-grade (0.7 dex scatter). (b) Higher-order cumulants beyond the mean-field Bernoulli variance, or mean-field-breaking tidal dynamics, may be required to account for the residual 0.34 dex $g_{\rm obs}$ scatter. (c) Ultra-faint dSph ($\Upsilon_{\rm dyn} > 100$) lie at the extrapolation limit of the current sample and warrant independent analysis.

8. Conclusions

We have presented a membrane cosmology framework with the following established results:

1. **Geometric mean law (A-grade):** $g_c = \eta\sqrt{a_0\, G\Sigma_0}$, $\alpha = 0.5$, ΔAIC $= -130$ vs MOND, 175 SPARC galaxies. Jackknife-verified (96.5%). Origin: deep-MOND algebraic consequence ($\alpha = 0.471 \pm 0.036$, $p(0.5) = 0.43$).

2. **MOND from first principles (A-grade):** Simple interpolation $\mu(x) = x/(1 + x)$ derived from elastic membrane potential equilibrium. Not an empirical ansatz.

3. **C15 final form (A-grade):** $g_c = 0.584\, \Upsilon_d^{-0.361} \sqrt{a_0\, v_{\rm flat}^2/h_R}$. $R^2 = 0.607$, scatter = 0.286 dex. MOND rejected at $p = 1.66 \times 10^{-53}$.

4. **C15 is minimal sufficient statistic (A-grade, v4.7):** Six $S_{\rm gal}$ models all rejected. All parameter additions yield LOO < 2%. 94% of intrinsic variance irreducible. Scatter = membrane fold diversity.

5. **$\kappa = 0$: membrane rigidity zero (A-grade, v4.7):** $\kappa \sim 10^{-40}$ kpc^2. $\mathcal{L} \approx U(\epsilon; c)$. Each radius obeys $\mu(x)$ independently. Four free parameters eliminated.

6. **C14 pattern is C15 bias projection (A-grade, v4.7):** Ratio $g_c^{\rm local}/g_c^{\rm global} = 1.035$. RMS = 0.090 dex. C14 dynamical formulation retracted.

7. **$F_{\rm conf}$ derivation (A-grade):** $-c\ln(1-\epsilon)$ from forest + cavity MF effective theory (Lemmas 1–4).

8. **Weak lensing $Q > 1$ in 9/9 clusters (A-grade):** Wilcoxon $p = 0.0020$. cl1: membrane ΔAIC $= -4.4$ vs NFW. Level B+.

9. **Deep-MOND: g_c independent of $\Sigma_{\rm bar}$ (A-grade):** $r(g_c, \Sigma_{\rm bar}) = 0.07$. Membrane responds to $\Sigma_{\rm dyn}$ ($r = 0.73$), not baryonic mass.

10. **BTFR slope is Simpson's paradox artifact (A-grade):** Within-bin slope $= 0.54 \pm 0.07$ confirms $\alpha = 0.5$ non-circularly. Dwarf $b_{gc} = 2.32$: membrane dominates rotation.

11. **Dwarf regime verified (A, v4.7.7):** SPARC-internal: 8 LITTLE THINGS dwarfs, scatter $= 0.232$ dex, KS $p = 0.67$. External: 11 non-SPARC LITTLE THINGS (Oh et al., 2015), scatter $= 0.289$ dex, bias $= -0.004$ dex, KS $p = 0.73$. C15 uniform across $v_{\rm flat} = 12$–125 km/s. Upgraded from B+ via independent confirmation.

12. **Morphological-type bridge (A-grade, v4.7):** T-type vs $S_{\rm gal}$: $\rho = +0.89$, $p = 4 \times 10^{-60}$. Conceptual correspondence; no additional predictive power.

13. **Weak lensing RAR (A grade, v4.7.6):** KiDS-1000 (Brouwer et al., 2021) constrains C15 at Mpc scales but is CGM-limited (B grade). Independent HSC-SSP Y3 + GAMA DR4 analysis (157,338 isolated lenses, 503M pairs, 3 fields) yields $g_c = 2.73 \pm 0.11\, a_0$, ΔAIC $= +472$ favouring C15 over MOND. g_c–M_* slope ($+0.166 \pm 0.041$) rejects MOND at 4σ. Slope budget: C15 ($+0.075$) + Model B ($+0.017$) + 2-halo ($+0.034$) $= +0.126$, reducing the initial 2.2σ tension to 0.98σ.

14. **Self-consistent equation and η_0 origin (A, v4.7.7):** Membrane self-consistent equation $s = 1/(1 + \exp(-\Delta U(sQ)/T_m))$ with $T_m = \sqrt{6}$ yields $\langle s \rangle = 0.58$, identifying $\eta_0 = 0.584$ as the mean elastic fraction. The 8.3% β discrepancy (-0.331 vs -0.361) is resolved as finite-T_m absorption of Υ_d dependence by $s(Q)$, requiring no parameter adjustment. $T_m = \sqrt{6}$ confirmed by scatter optimisation ($T_m^{\rm opt} = 2.35$, 0.24σ from $\sqrt{6}$). Υ_d independence verified by partial correlation: $r(g_c, \Upsilon_d | f_{\rm gas}) = -0.437$, $r(g_c, f_{\rm gas}) = +0.06$ (n.s.).

15. **J3 regime inversion (A, v4.7.8):** Direct C15 application to 31 dSph galaxies fails with $+1.5$ dex systematic bias (four model variants consistent), not absorbed by Υ_d transformation. Root cause: $M_{J,\rm mem} \ll M_{J,\rm std}$ in 28/31 dSph (vs $M_{J,\rm std} \leq M_{J,\rm mem}$ in SPARC), inverting the causal direction of membrane–baryon coupling. In the J3 regime, g_c is set by intrinsic membrane parameters (T_m, a_0), not by baryonic surface density.

16. **Bernoulli prediction $G_{\rm Strigari} = s_0(1 - s_0)\, a_0$ (A, v4.7.8):** The $c \to 0$ limit of $U(\epsilon; c)$ gives $s_0 = 1/(1 + \exp(3/(2T_m))) = 0.3515$ ($T_m = \sqrt{6}$). The thermal Bernoulli variance predicts $G_{\rm Strigari} = 0.228\, a_0$ ($= 2.74 \times 10^{-11}$ m/s^2). Verified in two independent populations: dSph 31-galaxy sample ($0.240\, a_0$, 5% agreement) and bridge galaxy outer points ($0.219\, a_0$, 4% agreement). Non-analytic $\sqrt{Q}$ correction $s(Q) \approx s_0 + 0.110\sqrt{Q}$ arises from the $\ln(1-\epsilon)$ term.

17. **Continuous C15 $\to$ Strigari transition (A, v4.7.8):** Four SPARC bridge galaxies (ESO444-G084, NGC2915, NGC1705, NGC3741) with outer $\Upsilon_{\rm dyn} > 10$ exhibit statistically significant radial increase of $\Upsilon_{\rm dyn}(R)$ (4/4 galaxies, $z = 2.4$ to 21.6). Mann–Whitney tests confirm outer $>$ inner Υ in 4/4 galaxies ($p < 0.03$). Bridge outer median $\Upsilon_{\rm dyn} = 12.76$ sits at the center of the J3 transition band (10–30). All four A-grade criteria satisfied: the membrane regime transitions continuously within individual galaxies.

18. **$g_{\rm obs}$ L_V-independence at machine precision (A, v4.7.8):** Multivariate regression $\log g_{\rm obs}$ vs $(\log\sigma, \log r_h, \log L_V)$ in 31 dSph galaxies yields $\beta_{L_V} = -2 \times 10^{-7}$ with $R^2 = 1.0000$, recovering $g_{\rm obs} = 3\sigma^2/r_h$ as an exact identity. The L_V contribution is at machine-precision level. Cross-verified by orthogonalized regression and partial correlation $r(g_{\rm obs}, L_V | \sigma, r_h) = -0.236$ ($p = 0.20$). The apparent slope -1 of $\log g_c$ vs $\log L_V$ is shown to be the definitional consequence

of $g_c = g_{\rm obs}^2/g_{\rm bar}$: regression yields $(\beta_\sigma, \beta_{r_h}, \beta_{L_V}) = (+3.82, +0.16, -1.07)$, all within 0.2 of the definitional prediction $(+4, 0, -1)$.

19. **Scatter decomposition and host-environment independence (A, v4.7.8):** Four-component decomposition of the 0.350 dex $g_{\rm obs}$ scatter around the Bernoulli prediction: measurement 0.062 dex (3.2%), host-between 0.095 dex (7.4%), physical residual 0.331 dex (89.4%). MW satellites ($N = 15$) and M31 satellites ($N = 11$) yield essentially identical mean offsets from $G_{\rm Strigari}$ (+0.030 vs +0.032 dex), establishing that $G_{\rm Strigari}$ is independent of the host galactic environment. Measurement contribution is negligible; the physical residual is dominated by the dSph σ–size relation $\sigma \propto r_h^{0.27}$.

Retracted/revised in v4.7.8: Condition 14 dynamical formulation $g_{c,\rm eff}(r)$ (v4.7); $\kappa \to A, B, \tau$ inverse estimation; Euler–Lagrange derivation pathway; gradient term $\kappa(d\epsilon/dr)^2$; $S_{\rm gal}$ predictive significance; H3 hysteresis-only hypothesis for dSph offset (rejected in v4.7.8 by three independent tests); strict Strigari universality ($g_{\rm obs}$ = const) — the r_h dependence is accommodated by the empirical σ–size relation, with $g_{\rm obs}$ $M_{\rm bar}$-independence retained as the membrane prediction.

Priority future work: (1) BIG-SPARC external verification ($\sim$ 4000 galaxies). (2) Non-SPARC LITTLE THINGS galaxies: full 3.6 μm photometry for precise Υ_d. (3) Ultra-faint dSph verification ($\Upsilon_{\rm dyn} > 100$ regime, $N > 50$ sample). (4) Rubin LSST / Euclid weak lensing (2-halo detection at 2–3σ). (5) Independent T_m verification from non-SPARC datasets. (6) Theoretical origin of the dSph σ–size relation $\sigma \propto r_h^{0.27}$: identify whether higher cumulants beyond the mean-field $s_0(1-s_0)$, or mean-field-breaking dynamical effects (tidal history, formation epoch), dominate the 0.34 dex $g_{\rm obs}$ scatter.

Acknowledgments

This research is based on observations made with ESO Telescopes at the La Silla Paranal Observatory under programme IDs 177.A-3016, 177.A-3017, 177.A-3018, and 179.A-2004, and on data products produced by the KiDS consortium. The KiDS production team acknowledges support from: Deutsche Forschungsgemeinschaft; ERC; NOVA and NWO-M grants; Target; the University of Padova; and the University Federico II (Naples).

The Hyper Suprime-Cam (HSC) collaboration includes the astronomical communities of Japan and Taiwan, and Princeton University. The HSC instrumentation and software were developed by the National Astronomical Observatory of Japan (NAOJ), the Kavli Institute for the Physics and Mathematics of the Universe (Kavli IPMU), the University of Tokyo, KEK, ASIAA, and Princeton University. Funding was contributed by the FIRST program from the Japanese Cabinet Office, MEXT, JSPS, JST, the Toray Science Foundation, NAOJ, Kavli IPMU, KEK, ASIAA, and Princeton University. GAMA is a joint European–Australasian project based around a spectroscopic campaign using the Anglo-Australian Telescope, funded by STFC (UK), ARC (Australia), AAO, and participating institutions.

References

Iu. V. Babyk, B. R. McNamara, P. E. J. Nulsen, H. R. Russell, et al. A Universal Entropy Profile for the Hot Atmospheres of Galaxies and Clusters. *The Astrophysical Journal*, 857:32, 2018. doi: 10.3847/1538-4357/aab3c9.

C. M. Brasseur, N. F. Martin, A. V. Maccio, H.-W. Rix, and X. Kang. Nature versus Nurture: Luminous Satellite Galaxies and the Chemical Evolution of Their Hosts. *The Astrophysical Journal*, 729:23, 2011. doi: 10.1088/0004-637X/729/1/23.

D. Brout, D. Scolnic, B. Popovic, A. G. Riess, et al. The Pantheon+ Analysis: Cosmological Constraints. *The Astrophysical Journal*, 938:110, 2022. doi: 10.3847/1538-4357/ac8e04.

M. M. Brouwer, V. Demchenko, J. Harnois-Déraps, M. Bilicki, et al. The weak gravitational lensing signal around galaxies in the KiDS survey: dependence on galaxy colour and Sersic index. *Astronomy & Astrophysics*, 650:A113, 2021. doi: 10.1051/0004-6361/202040164.

M. L. M. Collins, S. C. Chapman, R. M. Rich, R. A. Ibata, et al. A Kinematic Study of the Andromeda Dwarf Spheroidal System. *The Astrophysical Journal*, 768:172, 2013. doi: 10.1088/0004-637X/768/2/172.

R. Dalal, X. Li, A. Nicola, et al. Hyper Suprime-Cam Year 3 Results: Cosmology from cosmic shear power spectra. *Physical Review D*, 108:123519, 2023. doi: 10.1103/PhysRevD.108.123519.

D. A. Hunter, D. Ficut-Vicas, T. Ashley, E. Brinks, et al. Little Things. *The Astronomical Journal*, 144:134, 2012. doi: 10.1088/0004-6256/144/5/134.

D. A. Hunter, B. G. Elmegreen, S.-H. Oh, et al. The LITTLE THINGS Stellar Disks. *The Astronomical Journal*, 161:71, 2021. doi: 10.3847/1538-3881/abd089.

G. Iorio, F. Fraternali, C. Nipoti, E. Di Teodoro, J. I. Read, and G. Battaglia. LITTLE THINGS in 3D: robust determination of the circular velocity of dwarf irregular galaxies. *Monthly Notices of the Royal Astronomical Society*, 466:4159, 2017. doi: 10.1093/mnras/stw3285.

M. P. Julio et al. Weak lensing RAR. *Astronomy & Astrophysics*, 703:A106, 2025.

N. Kacharov, G. Battaglia, M. Rejkuba, et al. Stellar kinematics in isolated dwarfs. *Monthly Notices of the Royal Astronomical Society*, 466:2006, 2017. doi: 10.1093/mnras/stw3188.

F. Lelli, S. S. McGaugh, and J. M. Schombert. SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves. *The Astronomical Journal*, 152:157, 2016. doi: 10.3847/0004-6256/152/6/157.

G. F. Lewis, R. A. Ibata, S. C. Chapman, A. W. McConnachie, et al. Kinematics of the Andromeda System. *Monthly Notices of the Royal Astronomical Society*, 375:1364, 2007. doi: 10.1111/j.1365-2966.2006.11385.x.

X. Li, H. Miyatake, W. Luo, et al. HSC Year 3 Shape Catalogue: Image Simulation Calibration. *Publications of the Astronomical Society of Japan*, 74:421, 2022. doi: 10.1093/pasj/psac006.

R. Maartens and K. Koyama. Brane-World Gravity. *Living Reviews in Relativity*, 13:5, 2010. doi: 10.12942/lrr-2010-5.

R. Mandelbaum, F. Lanusse, A. Leauthaud, et al. HSC shear catalog calibration via image simulations. *Monthly Notices of the Royal Astronomical Society*, 481:3170, 2018. doi: 10.1093/mnras/sty2420.

A. W. McConnachie. The Observed Properties of Dwarf Galaxies in and around the Local Group. *The Astronomical Journal*, 144:4, 2012. doi: 10.1088/0004-6256/144/1/4.

S. S. McGaugh and J. M. Schombert. Color-Mass-to-Light-Ratio Relations for Disk Galaxies. *The Astronomical Journal*, 148:77, 2014. doi: 10.1088/0004-6256/148/5/77.

S. E. Meidt, E. Schinnerer, G. van de Ven, et al. Reconstructing the Stellar Mass Distributions of Galaxies Using S4G IRAC 3.6 and 4.5 Micron Images. *The Astrophysical Journal*, 788:144, 2014. doi: 10.1088/0004-637X/788/2/144.

M. Milgrom. A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis. *The Astrophysical Journal*, 270:365, 1983. doi: 10.1086/161130.

K. Miyaoka, N. Okabe, et al. Weak-lensing study of XMM cluster sample with HSC-SSP data. *Publications of the Astronomical Society of Japan*, 71:62, 2019. doi: 10.1093/pasj/psz037.

S. More, S. Sugiyama, H. Miyatake, et al. Hyper Suprime-Cam Year 3 Results: Measurements of Clustering of SDSS-BOSS Galaxies, Galaxy-Galaxy Lensing, and Cosmic Shear. *Physical Review D*, 108:123520, 2023. doi: 10.1103/PhysRevD.108.123520.

S.-H. Oh, D. A. Hunter, E. Brinks, et al. High-Resolution Mass Models of Dwarf Galaxies from LITTLE THINGS. *The Astronomical Journal*, 149:180, 2015. doi: 10.1088/0004-6256/149/6/180.

L. Randall and R. Sundrum. An Alternative to Compactification. *Physical Review Letters*, 83:3370, 1999. doi: 10.1103/PhysRevLett.83.4690.

M. M. Rau, R. Dalal, T. Zhang, et al. Weak lensing tomographic redshift distributions for HSC Y3. *Monthly Notices of the Royal Astronomical Society*, 524:5109, 2023. doi: 10.1093/mnras/stad1962.

V. C. Rubin and Jr. Ford, W. K. Rotational Properties of 21 Sc Galaxies with a Large Range of Luminosities and Radii. *The Astrophysical Journal*, 238:471, 1980. doi: 10.1086/158003.

J. M. Schombert, S. S. McGaugh, and F. Lelli. The Mass Discrepancy-Acceleration Relation in Early-Type Galaxies. *Monthly Notices of the Royal Astronomical Society*, 483:1496, 2019. doi: 10.1093/mnras/sty3203.

J. D. Simon. The Faintest Dwarf Galaxies. *Annual Review of Astronomy and Astrophysics*, 57:375, 2019. doi: 10.1146/annurev-astro-091918-104453.

J. D. Simon and M. Geha. The Kinematics of the Ultra-faint Milky Way Satellites. *The Astrophysical Journal*, 670:313, 2007. doi: 10.1086/521816.

C. Stone, S. Courteau, J.-C. Cuillandre, et al. PROBES: Photometry and Rotation Curve Observations from Extragalactic Surveys. *The Astrophysical Journal Supplement Series*, 256:33, 2021. doi: 10.3847/1538-4365/ac1195.

L. E. Strigari, J. S. Bullock, M. Kaplinghat, J. D. Simon, M. Geha, B. Willman, and M. G. Walker. A common mass scale for satellite galaxies of the Milky Way. *Nature*, 454:1096, 2008. doi: 10.1038/nature07222.

E. J. Tollerud, R. L. Beaton, M. C. Geha, et al. The SPLASH Survey: Spectroscopy of 15 M31 Dwarf Spheroidal Satellite Galaxies. *The Astrophysical Journal*, 752:45, 2012. doi: 10.1088/0004-637X/752/1/45.

R. B. Tully and J. R. Fisher. A new method of determining distances to galaxies. *Astronomy & Astrophysics*, 54:661, 1977.

J. Tumlinson, M. S. Peeples, and J. K. Werk. The Circumgalactic Medium. *Annual Review of Astronomy and Astrophysics*, 55:389, 2017. doi: 10.1146/annurev-astro-091916-055240.

E. P. Verlinde. Emergent Gravity and the Dark Universe. *SciPost Physics*, 2:016, 2017. doi: 10.21468/SciPostPhys.2.3.016.

M. G. Walker, M. Mateo, E. W. Olszewski, et al. A Universal Mass Profile for Dwarf Spheroidal Galaxies? *The Astrophysical Journal*, 704:1274, 2009. doi: 10.1088/0004-637X/704/2/1274.

J. Wolf, G. D. Martinez, J. S. Bullock, et al. Accurate masses for dispersion-supported galaxies. *Monthly Notices of the Royal Astronomical Society*, 406:1220, 2010. doi: 10.1111/j.1365-2966.2010.16753.x.

H.-X. Zhang, D. A. Hunter, B. G. Elmegreen, et al. Stellar Populations and Ages of LITTLE THINGS Dwarf Galaxies. *The Astronomical Journal*, 143:47, 2012. doi: 10.1088/0004-6256/143/2/47.